

```
losetup $OVERLAY /run/initramfs/<overlay-file-path>
sz=$(blockdev --getsz $BASE)
echo 0 $sz snapshot $BASE $OVERLAY p 8 | dmsetup create
live-rw
```

An obvious drawback to this approach is that you may use only a filesystem supported by Volume Manager. This, in particular, rules out network filesystems which can be useful in the context of LTSP and similar diskless environments.

Unions/aufs

Unions and, later, *aufs* were designed to offer a union of multiple filesystems. Each filesystem is stacked on top of the preceding ones. The search for a file starts with the top stack and goes down the stack. Each filesystem may be of a different type. There are user space implementations of *unions* that use the fuse kernel module.

Ubuntu switched from *aufs* to *overlays*. *Overlays* is simpler in that it restricts the union to just two filesystems—an upper layer and a lower layer. This module may be included in the 3.10 Linux kernel.

The *overlays* mechanism is superb for packing a large set of applications into a limited space by taking advantage of the compression-enabled filesystem, like *squashfs*. You can go back to the original state any time by just clearing the upperdir. In addition to being useful in a live USB, it is an excellent option for small devices for which the distribution of updates is controlled. 

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